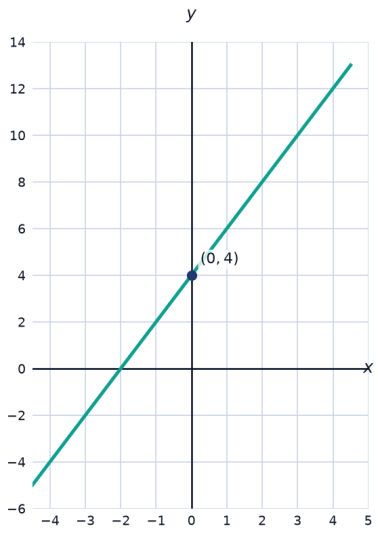
Sequences & Series

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

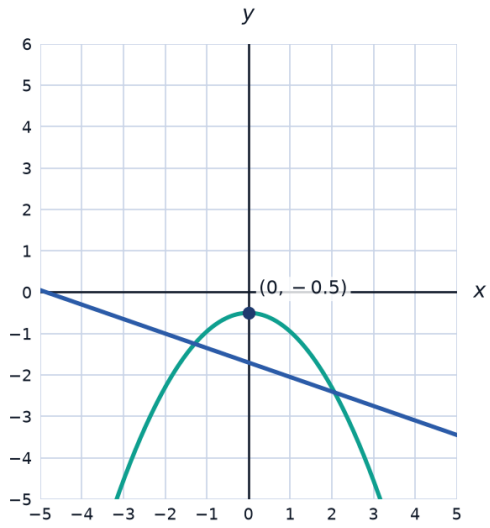
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 2x + 4$ in arithmetic series, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

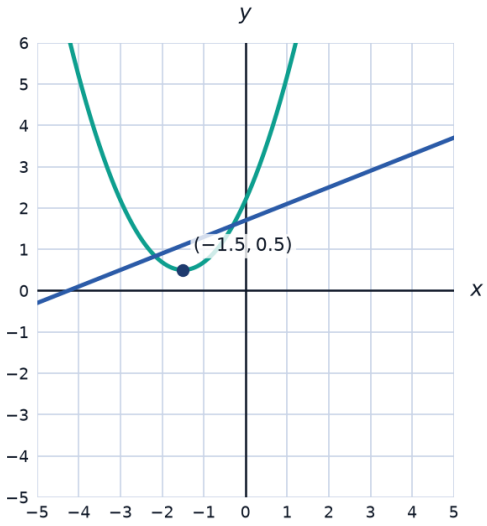
In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



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3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In ΔABC studying geometric series, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the sum of all integers between 10 and 100 (inclusive) that are divisible by 4.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In a geometric sequence, the 3rd term is 18 and the 6th term is 486. Find the common ratio r, the first term a_1, and the sum of the first 5 terms S_5.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

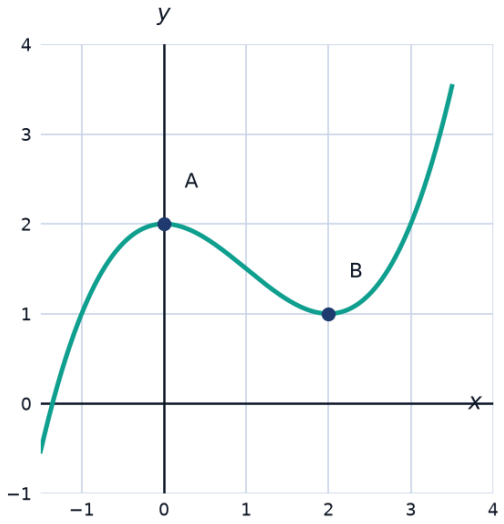
8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Sequences, Series, Induction & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.



10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An infinite geometric series has first term $a_1 = 36$ and converges to $S_\infty = 48$. What is the common ratio r, and what is the third term a_3?

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

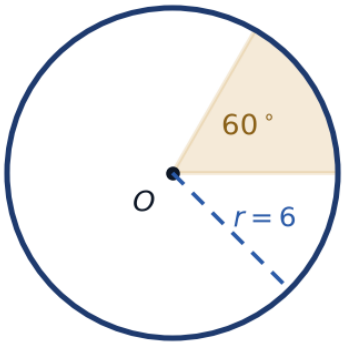
Evaluate the finite series $\sum_{k=1}^{4} (2k - 1)$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

An infinite geometric series has first term $a_1 = 5$ and common ratio $r = 0.6$. Find its sum S_∞.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

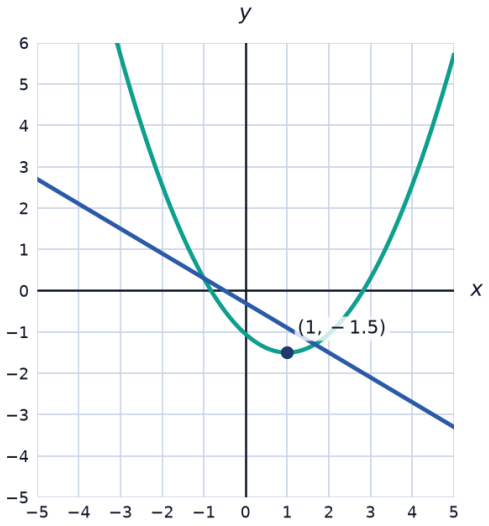
If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.



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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

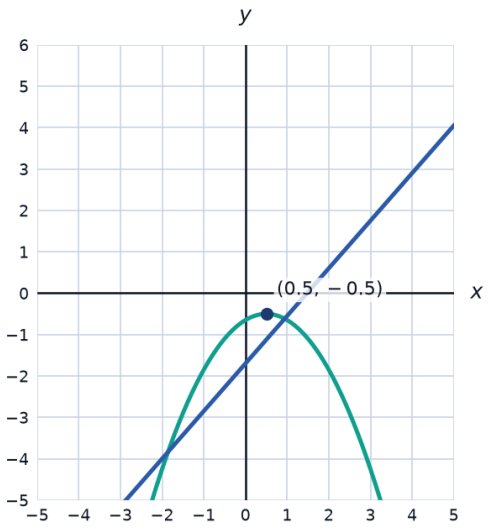
A polynomial $f(x)$ of degree 4 with real coefficients has leading coefficient 2 and zeros at $x = -1$ (multiplicity 2) and $x = 2 + i$. Write $f(x)$ in fully expanded standard form.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Probability & Statistics

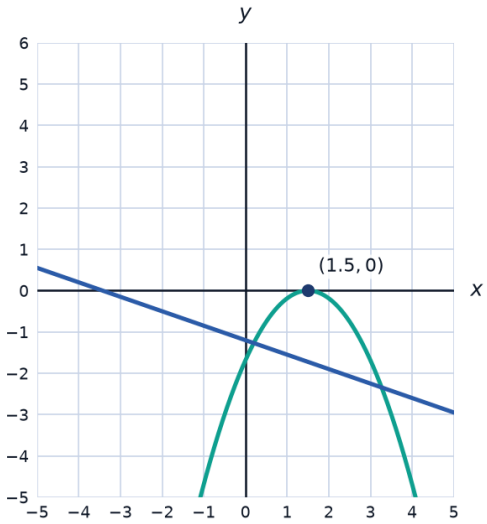
For what values of k does $x^2 - kx + 9 = 0$ have equal real roots?



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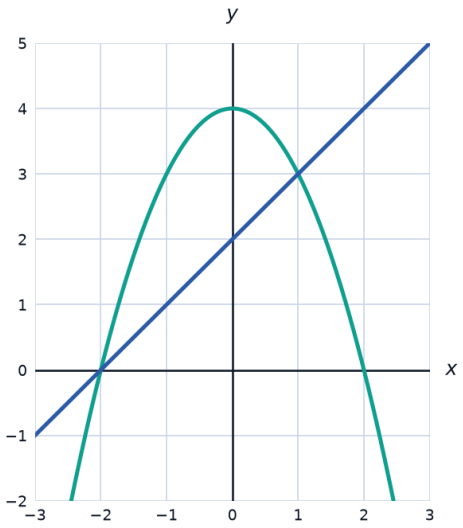
16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Polynomial Functions

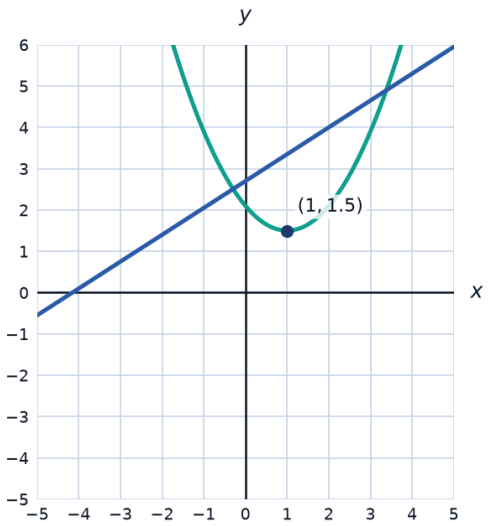
Find the average rate of change of the polynomial function $f(x) = 2x^3 - 5x + 4$ over the closed interval $[-1, 2]$.



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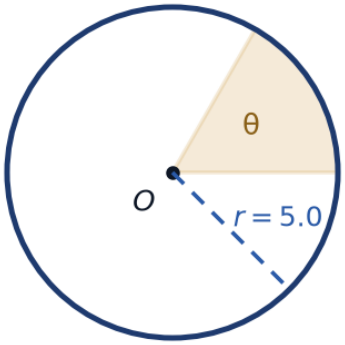
18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Rational Expressions

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).



19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.



20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Radicals & Rational Exponents

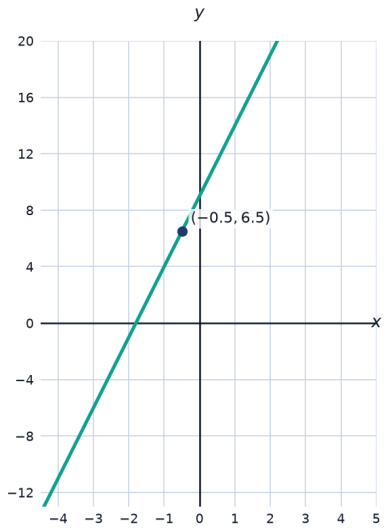
Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 9$ in notation for sequences and series, evaluate $f(2)$.



a. 19

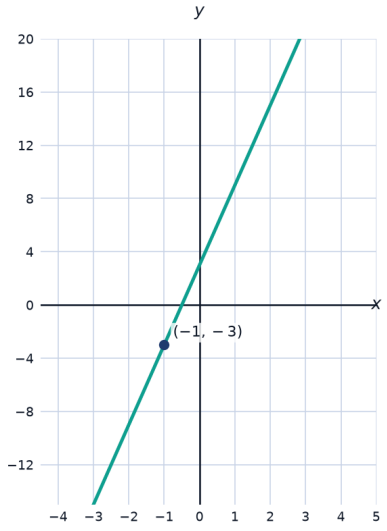
b. 24

c. 10

d. 23

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 3$ in arithmetics sequences, evaluate $f(2)$.



a. 15

b. 21

c. 12

d. 19

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for notation for sequences and series consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for arithmetics sequences consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for arithmetic series consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Prove: in any geometric sequence with positive terms, the geometric mean of two terms equals the intermediate terms.

a. Not always true

b. Only for $r = 1$

c. Use AM-GM

d. If a, ar, ar^2: $\sqrt{a \cdot ar^2} = ar =$ middle term

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $(a_n)_{n \geq 1}$ be an arithmetic sequence of real numbers with first term $a_1 > 0$ and non-zero common difference d. The terms a_1, a_3, and a_4 form the first three terms of a convergent geometric sequence $(g_k)_{k \geq 1}$, such that $g_1 = a_1$, $g_2 = a_3$, and $g_3 = a_4$. Let $S_\infty = \sum_{k=1}^{\infty} g_k$ denote the sum to infinity of the geometric sequence, and let $S_n = \sum_{j=1}^{n} a_j$ denote the n-th partial sum of the arithmetic sequence. If $M = \max_{n \geq 1} S_n$ is the maximum value attained by the partial sums of the arithmetic sequence, what is the value of $\frac{M}{S_\infty}$?

a. $\frac{5}{4}$

b. $\frac{81}{64}$

c. $\frac{9}{8}$

d. $\frac{5}{8}$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider an infinite geometric series with first term a and common ratio r, where $|r| < 1$: $a + ar + ar^2 + ar^3 + \ldots = 12$ If each term in the series is squared, the sum of the resulting infinite geometric series is: $a^2 + a^2r^2 + a^2r^4 + a^2r^6 + \ldots = 72$ What is the value of the common ratio, r?

a. $\frac{1}{3}$

b. $\frac{1}{2}$

c. $\frac{2}{3}$

d. $\frac{1}{4}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A geometric sequence has its first term a_1 and common ratio r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. Find the value of a_1 given that $req1$.

a. 1

b. 3

c. 7

d. 9

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An arithmetic sequence begins with the terms: $8, 13, 18, 23, \ldots$ What is the 15th term of the sequence?

a. 78

b. 83

c. 73

d. 117

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the arithmetic sequence where the first term $a_1 = 7$ and the common difference $d = -3$. Which of the following is the explicit formula for the nth term of this sequence?

a. $a_n = 7 + 3(n - 1)$

b. $a_n = 7 - 3(n - 1)$

c. $a_n = 3 - 7(n - 1)$

d. $a_n = -3 + 7(n - 1)$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A sequence is defined by the recurrence relation $a_n = 2a_{n-1} - a_{n-2} + 3$ for $n \geq 3$. If the first term $a_1 = 5$ and the second term $a_2 = 8$, find the value of a_4.

a. 17

b. 20

c. 23

d. 26

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $|r| < 1$, prove $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ using the limit of partial sums.

a. Use calculus only

b. Cannot be proved

c. $S_N = \frac{1 - r^{N+1}}{1 - r} \to \frac{1}{1-r}$ as $r^{N+1} \to 0$

d. Only for $0 < r < 1$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Prove $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$ using the formula for arithmetic sums.

a. Cannot be proved simply

b. By FTA

c. By induction: base $n = 1$ trivially true; induction step involves algebraic manipulation

d. Use telescoping

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find a_1 and d for the arithmetic sequence where $S_{10} = 155$ and $a_{10} = 28$.

a. $a_1 = 3,\ d = \frac{25}{9}$

b. $a_1 = 5,\ d = 3$

c. $a_1 = 1,\ d = 3$

d. $a_1 = 3,\ d = 3$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Prove by induction: $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

a. Not provable

b. Only for even n

c. Use calculus

d. Base: $n = 1$: $1 = 6/6$. ✓. Step: add $(n+1)^2$ and factor.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the closed form for F_n (Fibonacci) using Binet's formula.

a. $F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$ where $= 1 + 52,\ = 1 - 52$

b. $F_n = \phi^{n-1}$

c. $F_n = 2^n/5$

d. $F_n = n\phi$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Rational Expressions

Using partial fractions, find $\sum_{k=1}^{n} \frac{1}{(2k-1)(2k+1)}$.

a. 1

b. $\frac{1}{2}$

c. $\frac{n}{2n-1}$

d. $\frac{n}{2n+1}$

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The formula for the sum of n terms of an arithmetic series is:

a. $S_n = \frac{a_1}{1-r}$

b. $S_n = \frac{n}{2}(a_1 + a_n)$

c. $S_n = na_1$

d. $S_n = a_1 \cdot \frac{r^n - 1}{r - 1}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Radicals & Rational Exponents

A geometric series with $|r| < 1$ has an infinite sum of:

a. $S = a_1 \cdot r$

b. $S = \frac{a_1}{1-r}$

c. Undefined

d. $S = \frac{a_1}{r}$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Conic Sections

Find the first term and common difference of the AP whose pth term is $3p + 2$.

a. $a_1 = 2, d = 3$

b. $a_1 = 5, d = 3$

c. $a_1 = 5, d = 2$

d. $a_1 = 3, d = 2$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Trigonometric Functions

Use sigma notation for $1^2 + 2^2 + 3^2 + \cdots + n^2$.

a. $\prod_{k=1}^{n} k^2$

b. $\sum_{k=0}^{n} k^2$

c. $\sum_{k=1}^{n} k^2$

d. n^2

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Functions & Their Graphs

Insert 3 arithmetic means between 5 and 21.

a. $7, 11, 15$

b. $9, 13, 17$

c. $8, 12, 16$

d. $10, 14, 18$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Matrices & Systems

Is the series $1 - 1 + 1 - 1 + \cdots$ convergent?

a. Yes, sum = 1

b. Yes, sum = 0

c. No (it diverges)

d. Yes, sum = 1/2

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Probability & Statistics

The sum of the first n squares: $\sum_{k=1}^{n} k^2 =$

a. $\frac{n(n+1)}{2}$

b. n^3

c. $\frac{n^2(n+1)}{2}$

d. $\frac{n(n+1)(2n+1)}{6}$

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